# SpinFlow: A Physics-Informed Spin Field Framework for Traffic Phase Inference and Transition Detection

Haopeng Deng, Fucheng Zheng, and Xinhai Xia

*Abstract*— **Active traffic management (ATM) is frequently hindered by traditional macroscopic models and rigid empirical thresholds that fail to capture metastable phase precursors, resulting in delayed, reactive interventions. To address this, we propose SpinFlow, a physics-informed spin-field framework unifying Kerner's three-phase theory with statistical physics for continuous macroscopic traffic phase inference. Inspired by the Heisenberg model, SpinFlow parametrizes spatially varying phase weights via a latent spin vector and a competitive-equilibrium mapping, allowing synchronized flow to emerge naturally. A physics-regularized Expectation-Maximization algorithm inverts this latent structure from high-resolution trajectories, jointly optimizing the spin field while softly enforcing mass conservation and spatial smoothness. We introduce the Phase Equilibrium Degree (PED) to quantify structural alignment and topologically localize phase-transition points. Across four real-world trajectory datasets, SpinFlow achieves $R_q^2$ up to 0.940, PED drops of 94.9–100 %, and interpretable phase maps that outperform three heterogeneous baselines on forward accuracy, physics consistency, and bottleneck localization. SpinFlow pinpoints congestion nucleation without prior network topology, yielding a data-driven, physics-consistent trigger for ATM.**

**Index Terms—**Traffic phase inference; three-phase traffic theory; fundamental diagram; physics-informed learning.

## I. INTRODUCTION

Proactive macroscopic traffic phase inference and transition identification are essential for enhancing operational efficiency and safety in Intelligent Transportation Systems (ITS). Classical macroscopic traffic flow theory originated from the fundamental diagram (FD) for traffic flow, which has evolved over nearly 90 years [1]—e.g., Greenshields' linear model and logarithmic or exponential variants—establishing the foundational density-flow-speed relationships. These inspired the Lighthill-Whitham-Richards (LWR) kinematic wave model [2], [3] that treats traffic as a compressible fluid obeying conservation laws. To ease the computational burden of the resulting PDEs, Daganzo proposed the Cell Transmission Model (CTM) [4]; second-order models such as Payne-Whitham (PW) [5] and Aw-Rascle-Zhang (AWZ) [6], [7] add velocity dynamics to capture non-equilibrium waves. Despite their maturity and utility, these two-phase (free vs. congested) frameworks with a single FD treat complex traffic dynamics isotropically. Consequently, they lack the endogenous mechanisms required to intrinsically explain key empirical phenomena—including hysteresis, capacity drop, multi-stability, and the spontaneous, nonlinear jumps between free and congested flow—owing to their strict reliance on continuous-medium and static-equilibrium assumptions.

To address these limitations, Kerner introduced three-phase traffic theory [8], [9], distinguishing free flow (F), synchronized flow (S), and wide moving jam (J). The theory characterizes the F→S transition as a first-order phase change, with wide moving jams nucleating internally within synchronized flow; the transition zone exhibits pronounced metastability, stochastic triggering, and spatiotemporal competition [10], [11], [12]. Empirical validations using trajectory data have extensively confirmed these features at geometric bottlenecks [13], [14]. Complementing these empirical advances, the statistical-physics perspective frames traffic as an emergent many-particle system [15], [16]. Nagatani's lattice hydrodynamic models [17], [18] revealed jamming transitions and critical phenomena. Notably, the structural isomorphism between traffic phase competition and lattice spin systems provides a particularly powerful analytical lens. While the Ising model [19] captures binary discrete states, the Heisenberg model [20] extends this to continuous three-dimensional vector representations, effectively overcoming the theoretical bottlenecks of hard-boundary classifications. Together with Boltzmann statistics [21] for phase probabilities and Landau theory [22] for order-parameter stability boundaries, these physics-inspired paradigms offer a rigorous mathematical foundation. The Expectation-Maximization (EM) algorithm [23] and its variational free-energy interpretation [24] provide the computational engine for such latent-variable inversion. While recent trajectory-driven calibrations [25] and hybrid physics-informed machine learning (PIML) [26], [27] have improved modeling robustness under sparse data, their reliance on neural backbones often sacrifices the strict interpretability inherent to fully transparent white-box models.

Despite these substantial advances, two critical gaps, persisting over 20 years since Kerner, severely limit engineering deployment and ATM. First, a persistent disconnection remains between microscopic phase-transition mechanisms and macroscopic, computable states. Existing macroscopic models and three-phase simulations excel at post-hoc reproduction but rarely translate complex latent physics into interpretable, spatially continuous phase representations. Second, the quantitative identification of critical equilibrium states remains elusive. Conventional state-recognition methodologies predominantly rely on hard empirical thresholds (e.g., rigid velocity or density cutoffs), which inherently fail to capture the subtle pre-transition precursors, structural entropy, and

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metastable fluctuations characterizing mixed-phase regimes. Consequently, modern control systems often intervene only after a shockwave has fully formed, entirely missing the narrow, low-cost window for preventive interventions.

This paper presents SpinFlow, a physics-informed spin-field framework that unifies Kerner's three-phase theory with statistical-physics latent-variable modeling for continuous macroscopic traffic phase inference and transition identification. Inspired by the Heisenberg model's continuous symmetry, SpinFlow parametrizes spatially varying phase weights $\boldsymbol{\pi}(x)$ via a latent spin vector $\mathbf{s}(x) \in \mathbb{R}^3$ and a competitive-equilibrium mapping. Free-jam competition bias and strength drive this mapping, allowing synchronized flow to emerge naturally as the balanced equilibrium outcome. A physics-regularized EM algorithm inverts the latent structure from high-resolution trajectory data [28], [29], [30]. Finally, the framework introduces the Phase Equilibrium Degree (PED) to quantify structural alignment and to topologically localize quasi-equilibrium regions and phase-transition points, providing a data-driven, physics-consistent understanding of congestion nucleation and a computable trigger for ATM.

Experiments show tight in-sample forward consistency and interpretable phase maps; detected transitions are consistent with documented bottleneck features. A companion open-source repository is publicly maintained at `https://github.com/QianyHP/spinflow-traffic` under the MIT License, including evaluation scripts, visualization utilities, and extended analyses to support reproducibility. The remainder of the paper is organized as follows: Sec. II details observations and phase prototypes; Sec. III presents the SpinFlow model; Sec. IV describes the EM inference and PED-based detection; Sec. V reports experimental validation; and Sec. VI concludes with future directions.

## II. Observations and Phase Prototypes

### A. Trajectory Representation and Edie Macroscopic Variables

Vehicle trajectories $\mathcal{Z}$ collected on a road segment $x \in [0, L]$ over $t \in [t_0, t_0 + \Delta T]$ form the basis of our analysis. For the $n$th trajectory $x_n(t)$ with speed $u_n(t) = \dot{x}_n(t)$ and any spatiotemporal region $\mathcal{A} \subset \mathbb{R}^2$, the total traveled distance and total time spent are

$$d(\mathcal{A}) = \sum_n d_n(\mathcal{A}), \qquad \tau(\mathcal{A}) = \sum_n \tau_n(\mathcal{A}), \tag{1}$$

where $d_n(\mathcal{A})$ and $\tau_n(\mathcal{A})$ denote the spatial and temporal projection lengths of trajectory $n$ inside $\mathcal{A}$. Edie's macroscopic variables [31] follow as

$$\rho(\mathcal{A}) = \frac{\tau(\mathcal{A})}{|\mathcal{A}|}, \qquad q(\mathcal{A}) = \frac{d(\mathcal{A})}{|\mathcal{A}|}, \qquad v(\mathcal{A}) = \frac{q(\mathcal{A})}{\rho(\mathcal{A})}, \tag{2}$$

with $|\mathcal{A}|$ representing the area of $\mathcal{A}$; reported values use $\rho$ in veh/km, $q$ in veh/h, and $v$ in km/h. Since large rectangular windows obscure transient waves, eq. (2) is evaluated on sampled local regions that approximate quasi-stationarity.

### B. Quasi-Stationary Sampling via Parallelograms

Building on the approach of He and Wu [25], sampling employs parallelograms $\{\mathcal{P}_p\}_{p=1}^N$ in the $(t, x)$ plane to capture local homogeneity. To capture wave propagation, the geometry of $\mathcal{P}_p$ is dynamically adapted: long edges align with the congestion wave speed while short edges follow the target speed $v^*$, ensuring trajectories inside $\mathcal{P}_p$ are nearly parallel.

Computation for each candidate $\mathcal{P}_p$ yields

$$(\rho_p, q_p, v_p) = \big(\rho(\mathcal{P}_p), q(\mathcal{P}_p), v(\mathcal{P}_p)\big), \tag{3}$$

recorded at the spatial center $x_p$. This process generates the observation vector

$$\mathbf{y}_p = (\rho_p, q_p, v_p, x_p). \tag{4}$$

A stability metric evaluates each region by penalizing internal speed fluctuations and deviations from $v^*$, given by

$$\mathrm{Score}(\mathcal{P}_p) = w_{\mathrm{CV}} \cdot \mathrm{CV}(\mathcal{P}_p) + w_{\mathrm{NAE}} \cdot \mathrm{NAE}(\mathcal{P}_p), \tag{5}$$

where $\mathrm{CV}(\mathcal{P}_p)$ is the coefficient of variation of speeds and $\mathrm{NAE}(\mathcal{P}_p)$ is the normalized absolute error relative to $v^*$. An Otsu threshold $\eta$ [32], derived from the empirical score distribution with interquartile range IQR, maps scores to continuous quality weights via

$$w_p = \frac{1}{1 + \exp\big(\beta_w(\mathrm{Score}(\mathcal{P}_p) - \eta)\big)}, \qquad \beta_w = \ln(9)/\mathrm{IQR}. \tag{6}$$

This logistic weighting suppresses low-quality regions ($w_p \approx 0$) during prototype fitting and inference. Fig. 1 contrasts the parallelogram sampling with traditional rectangular sampling, showing the reduction in intra-window velocity variance.

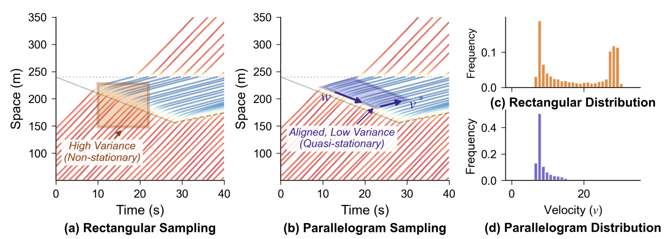


Fig. 1. (a) Rectangular window straddles the shockwave: bimodal velocity distribution (high $\sigma^2$). (b) Parallelogram aligned with wave speed $w$ and vehicle speed $v^*$: quasi-stationary region (low $\sigma^2$). (c, d) Velocity histograms.

### C. Three-Phase Prototype Fundamental Diagrams

Three phase prototypes $\mathcal{G} = \{F, S, J\}$ represent free flow (F), synchronized flow (S), and wide moving jam (J), respectively. Each phase $g \in \mathcal{G}$ follows a triangular FD in the sense of Newell's kinematic wave theory [33]:

$$q_g(\rho) = \min\Big\{v_f^{(g)}\rho,\ Q^{(g)},\ |w^{(g)}|\big(\rho_{\mathrm{jam}}^{(g)} - \rho\big)\Big\}, \qquad \rho \in [0, \rho_{\mathrm{jam}}^{(g)}]. \tag{7}$$

The parameter vector $\boldsymbol{\theta}_g = (v_f^{(g)}, w^{(g)}, \rho_{\mathrm{jam}}^{(g)}, Q^{(g)})$, where $Q^{(g)}$ is capacity, respects the constraints $v_f^{(g)} > 0$, $w^{(g)} < 0$, $\rho_{\mathrm{jam}}^{(g)} > 0$, and $Q^{(g)} > 0$. Phase attractors, identified by K-Means clustering in the normalized $(\rho, v)$ space, partition observations into clusters $\{\mathcal{D}_g\}_{g\in\mathcal{G}}$. Weighted least squares then fits each prototype,

$$\boldsymbol{\theta}_g^* = \arg\min_{\boldsymbol{\theta}_g} \sum_{p\in\mathcal{D}_g} \omega_p \Big(q_g(\rho_p;\boldsymbol{\theta}_g) - q_p\Big)^2, \tag{8}$$

where $\omega_p = w_p$ denotes the quasi-stationary weight from (6). These fitted prototypes form the phase-conditioned basis for the spatial mixture model.

## III. SpinFlow Model: Latent Spin Field and Competitive-Equilibrium Mapping

### A. Spatial Mixture FD and Latent Spin Field

Spatial heterogeneity is modeled by mixing the three phase prototypes with position-dependent weights $\boldsymbol{\pi}(x) = (\pi_F(x), \pi_S(x), \pi_J(x))$. The spatial mixture FD for density $\rho$ at location $x$ becomes

$$q(\rho; x) = \sum_{g\in\mathcal{G}} \pi_g(x)\, q_g(\rho), \tag{9}$$

subject to $\pi_g(x) \geq 0$ and $\sum_{g\in\mathcal{G}} \pi_g(x) = 1$. Speed from the mixture follows implicitly as

$$v(\rho; x) = \begin{cases} \dfrac{q(\rho;x)}{\rho}, & \rho > 0, \\ \sum_{g\in\mathcal{G}} \pi_g(x)\, v_f^{(g)}, & \rho = 0. \end{cases} \tag{10}$$

A latent spin field $\mathbf{s}(x) = (s_x(x), s_y(x), s_z(x)) \in \mathbb{R}^3$ parametrizes $\boldsymbol{\pi}(x)$. This spin vector offers a continuous representation of phase attractors while permitting mixed states in transition regions. Crucially, the magnitude $|\mathbf{s}(x)|$ remains unconstrained; the inversion process increases $|\mathbf{s}(x)|$ where data support a deterministic phase and keeps $|\mathbf{s}(x)|$ small where evidence is diffuse. We refer to this unconstrained spin magnitude as *Phase Unfolding*.

### B. Competitive Mapping from Spin to Phase Weights

We construct the mapping from $\mathbf{s}(x)$ to $\boldsymbol{\pi}(x)$ using three preference scores that explicitly encode the competition between free and jam phases. These scores are defined as

$$\begin{aligned} h_F(\mathbf{s}(x)) &= s_z(x) - s_y(x), \\ h_J(\mathbf{s}(x)) &= s_y(x) - s_z(x), \\ h_S(\mathbf{s}(x)) &= s_x(x) - |s_z(x) - s_y(x)|. \end{aligned} \tag{11}$$

The scores $h_F$ and $h_J$ form an antisymmetric pair ($h_J = -h_F$), driving the primary free–jam dichotomy. In contrast, $h_S$ incorporates a penalty term $|s_z - s_y|$, suppressing phase S when the free–jam competition is fierce. A Boltzmann (Softmax) function then generates the weights

$$\pi_g(x) = \frac{\exp\big(\beta\, h_g(\mathbf{s}(x))\big)}{\sum_{g'\in\mathcal{G}} \exp\big(\beta\, h_{g'}(\mathbf{s}(x))\big)}, \qquad g \in \mathcal{G}, \tag{12}$$

where $\beta > 0$ is the inverse temperature (distinct from $\beta_w$ in (6)). This construction enforces the probability constraints on $\boldsymbol{\pi}(x)$ by design.

### C. Physical Interpretation: Bias and Competition Strength

The spin components admit a physical interpretation through the competition bias and competition strength,

$$b(x) = s_z(x) - s_y(x), \qquad \chi(x) = |b(x)|. \tag{13}$$

Substituting these into (11) yields

$$h_F = b, \qquad h_J = -b, \qquad h_S = s_x - \chi. \tag{14}$$

The sign of $b(x)$ dictates whether $F$ or $J$ dominates: $b(x) \gg 0$ implies $h_F \gg 0$ ($\pi_F(x) \approx 1$), whereas $b(x) \ll 0$ implies $h_J \gg 0$ ($\pi_J(x) \approx 1$). Phase S is favored when free–jam competition is nearly balanced. As $b(x) \to 0$, $\chi(x)$ vanishes and $h_S \approx s_x$. If $s_x(x) > 0$, $\pi_S(x)$ dominates and synchronized flow emerges from competitive equilibrium. Fig. 2 summarizes this spin-to-phase pipeline.

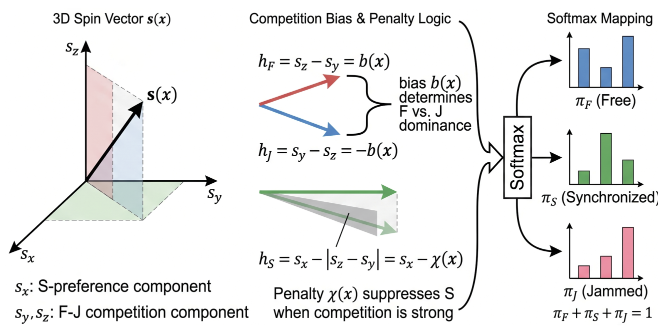


Fig. 2. Competitive-equilibrium mapping: bias $b(x)$ and penalty $\chi(x)$ govern preference scores $h_g$, which softmax maps to phase weights $\boldsymbol{\pi}(x)$.

## IV. Physics-Informed EM Inference and Transition Detection

### A. Generative Model and EM Responsibilities

Each FD observation point $\mathbf{y}_p = (\rho_p, q_p, v_p, x_p)$ is modeled as a sample drawn from a mixture with latent phase label $z_p \in \mathcal{G} = \{F, S, J\}$. Conditional on $z_p = g$, the flow $q_p$ is assumed to follow a Gaussian distribution centered at the prototype $q_g(\rho_p;\boldsymbol{\theta}_g)$,

$$q_p = q_g(\rho_p;\boldsymbol{\theta}_g) + \varepsilon_{q,p}, \qquad \varepsilon_{q,p} \sim \mathcal{N}(0, \sigma_q^2). \tag{15}$$

Marginalizing over $z_p$ with the spatial mixture weights $\pi_g(x_p)$ yields the likelihood

$$p(q_p \mid \rho_p, x_p) = \sum_{g\in\mathcal{G}} \pi_g(x_p)\, \mathcal{N}\big(q_p;\, q_g(\rho_p;\boldsymbol{\theta}_g),\, \sigma_q^2\big). \tag{16}$$

Neal and Hinton [24] showed that EM minimizes a variational free energy, under which the E-step is the optimal posterior. The E-step at iteration $t$ computes the posterior responsibilities

$$\begin{aligned} \gamma_{p,g}^{(t)} &= \Pr\big(z_p = g \mid q_p, \rho_p, x_p; \mathbf{s}^{(t)}, \boldsymbol{\theta}^{(t)}\big) \\ &= \frac{\pi_g^{(t)}(x_p)\, \mathcal{N}\big(q_p;\, q_g(\rho_p;\boldsymbol{\theta}_g^{(t)}),\, \sigma_q^2\big)}{\sum_{g'\in\mathcal{G}} \pi_{g'}^{(t)}(x_p)\, \mathcal{N}\big(q_p;\, q_{g'}(\rho_p;\boldsymbol{\theta}_{g'}^{(t)}),\, \sigma_q^2\big)}. \end{aligned} \tag{17}$$

**Algorithm 1** EM-Based Joint Spin–FD Inference

**Require:** Observations $\{(\rho_p, q_p, x_p, w_p)\}_{p=1}^{N}$; prototype FDs $\{q_g(\cdot;\boldsymbol{\theta}_g)\}_{g\in\mathcal{G}}$; grid $\{x_i\}_{i=1}^{M}$
**Require:** Hyperparameters $\lambda_{\text{phys}}, \lambda_{\text{smooth}}, \sigma, \beta$
**Ensure:** Spin field $\mathbf{s}(x_i)$ and mixture weights $\boldsymbol{\pi}(x_i)$

Initialize $\boldsymbol{\theta}_g^{(0)}$ and $\mathbf{s}^{(0)}(x_i)$
**for** $t = 0, 1, \ldots$ until convergence **do**
  $\boldsymbol{\pi}^{(t)}(x_i) \leftarrow \Pi\big(\mathbf{s}^{(t)}(x_i)\big)$ via (12)
  **E-step:** compute $\gamma_{p,g}^{(t)}$ via (17)
  aggregate $\pi_{g,\text{tar}}^{(t)}(x_i)$ via (18)
  **M-step (FD):** update $\boldsymbol{\theta}_g^{(t+1)}$ via (19)
  **M-step (spin):** update $\mathbf{s}^{(t+1)}$ by minimizing (20)
**end for**
**return** $\mathbf{s}(x_i)$ and $\boldsymbol{\pi}(x_i)$

**Algorithm 2** PED-Based Phase-Transition Detection

**Require:** $\boldsymbol{\pi}(x_i)$, observed $q_{\text{obs}}$, $\rho_{\text{obs}}$ from Alg. 1; threshold $\tau_H = \ln 2$; margin $\delta$
**Ensure:** Primary site $x^*$; profile $\{\text{PED}(x_i)\}$; metric $\Delta\text{PED}$

$\forall\, x_i$: $H(x_i) \leftarrow -\sum_{g\in\mathcal{G}} \pi_g \log \pi_g$
  and $\text{PED}(x_i) \leftarrow \exp\big(-\Delta\mathcal{F}(x_i)\big)$ via (21)–(22)
$\Omega_Q \leftarrow \{x_i \in [\delta, L-\delta] \mid H(x_i) \geq \tau_H\}$ via (23)
**if** $\Omega_Q \neq \emptyset$ **then**
  $x^* \leftarrow \arg\min_{x\in\Omega_Q} \text{PED}(x)$ via (24)
**else**
  $x^* \leftarrow \arg\max_{x\in[\delta,\, L-\delta]} |\nabla_x \rho_{\text{obs}}|$ ▷ density-gradient fallback
**end if**
$\Delta\text{PED} \leftarrow 1 - \text{PED}(x^*)\big/\text{mean}\big\{\text{PED}(x_i) \mid x_i \notin \Omega_Q\big\}$
**return** $x^*$, $\{\text{PED}(x_i)\}$, $\Delta\text{PED}$

Gaussian kernel aggregation then converts these point-wise responsibilities into a continuous spatial target mixture,

$$\pi_{g,\text{tar}}^{(t)}(x) = \frac{\sum_{p=1}^{N} w_p\, K(x, x_p; \sigma)\, \gamma_{p,g}^{(t)}}{\sum_{p=1}^{N} w_p\, K(x, x_p; \sigma)}, \qquad K(x, x_p; \sigma) = \exp\left(-\frac{(x - x_p)^2}{2\sigma^2}\right), \tag{18}$$

weighted by the quasi-stationary score $w_p$.

### B. Spatial Aggregation and Regularized Spin Update

The M-step updates the prototype parameters via weighted least squares,

$$\boldsymbol{\theta}_g^{(t+1)} = \arg\min_{\boldsymbol{\theta}_g} \sum_{p=1}^{N} \gamma_{p,g}^{(t)}\, w_p \Big(q_g(\rho_p; \boldsymbol{\theta}_g) - q_p\Big)^2, \tag{19}$$

subject to feasibility constraints in Sec. II. Simultaneously, the spin field update aligns the induced mixture $\boldsymbol{\pi}(x) = \Pi(\mathbf{s}(x))$ with the aggregated target $\boldsymbol{\pi}_{\text{tar}}^{(t)}(x)$. The spin field $\mathbf{s}(x)$ evolves to minimize a regularized free energy functional that balances data consistency against physical conservation laws:

$$\begin{aligned}\mathbf{s}^{(t+1)} = \arg\min_{\mathbf{s}} \int_0^L \Big[ &-\sum_{g\in\mathcal{G}} \pi_{g,\text{tar}}^{(t)}(x) \log \pi_g(x) \\ &+ \lambda_{\text{phys}} (\partial_t \rho + \partial_x q)^2 + \lambda_{\text{smooth}} \|\partial_x \mathbf{s}\|_2^2 \Big] dx.\end{aligned} \tag{20}$$

where $q(\rho; x)$ denotes the mixture FD. Phase Unfolding lets $|\mathbf{s}(x)|$ grow where phase evidence is sharp and stay small in mixed regimes (no fixed radius on $\|\mathbf{s}\|$). The complete procedure is summarized in Algorithm 1.

### C. Phase Equilibrium Degree and Transition Detection

We quantify local mixture equilibrium by the mismatch between the EM-aggregated target $\boldsymbol{\pi}_{\text{tar}}(x)$ and the inferred mixture $\boldsymbol{\pi}(x)$. Defining the free-energy gap as

$$\begin{aligned}\Delta\mathcal{F}(x) &= D_{\text{KL}}\big(\boldsymbol{\pi}_{\text{tar}}(x) \,\|\, \boldsymbol{\pi}(x)\big) \\ &= \sum_{g\in\mathcal{G}} \pi_{g,\text{tar}}(x) \log \frac{\pi_{g,\text{tar}}(x)}{\pi_g(x)},\end{aligned} \tag{21}$$

we introduce the Phase Equilibrium Degree (PED):

$$\text{PED}(x) = \exp\big(-\Delta\mathcal{F}(x)\big) \in (0, 1]. \tag{22}$$

The phase-coexistence window $\Omega_Q$ is formed by thresholding the Shannon entropy $H(x)$ of $\boldsymbol{\pi}(x)$:

$$\Omega_Q = \{x \mid H(x) \geq \tau_H = \ln 2\}, \tag{23}$$

where $\tau_H = \ln 2$ is the two-phase equal-weight boundary. The *primary bottleneck site* is then defined as

$$x^* = \arg\min_{x\in\Omega_Q} \text{PED}(x), \tag{24}$$

the cell at which the model–data equilibrium is most severely broken. $x^*$ identifies the nucleation center of the transition and represents the highest-priority location for ATM intervention. The full detection procedure is given in Algorithm 2.

## V. Experimental Validation

### A. Datasets and Setup

Validation uses four real-world trajectory datasets: YTDJ and RML from the Ubiquitous Traffic Eyes (UTE, China) [28], HighD-58 (Germany) [29], and NGSIM I-80 (USA) [30], covering expressway bottlenecks, on-ramps, and a highway section. All datasets provide per-vehicle position and speed at dataset-specific sampling rates; Fig. 3 shows their observed spacetime speed patterns. Table I gives key statistics.

TABLE I
Dataset statistics. FPS: sampling rate; $N_{\text{veh}}$: vehicles in the analysis window; $\bar{v}$: mean speed; FD pts: sampled observations.

| Scenario | Length (m) | Duration (s) | FPS | $N_{\text{veh}}$ | Lanes | $\bar{v}$ (km/h) | FD pts |
|---|---|---|---|---|---|---|---|
| UTE-YTDJ | 362 | 545 | 24 | 1072 | 5 | 17.2 | 512 |
| UTE-RML | 220 | 300 | 30 | 555 | 4 | 15.2 | 564 |
| HighD-58 | 425 | 389 | 25 | 373 | 4 | 31.0 | 161 |
| NGSIM I-80 | 540 | 800 | 10 | 1634 | 5 | 25.0 | 232 |

For each scenario, FD points are sampled following Sec. II. The grid uses $\Delta x = 2\,\text{m}$. All models share the same FD points, spatial grid, and seeds. Experiments are repeated over five seeds (42, 137, 256, 314, 628). Statistical significance is assessed by paired $t$-tests on $\text{RMSE}_q$ (SpinFlow vs. alternatives, two-sided). Key hyperparameters: $\lambda_{\text{phys}} = 0.1$, $\lambda_{\text{smooth}} = 0.02$, $\lambda_{\text{fd}} = 1.0$, learning rate $= 0.05$, EM

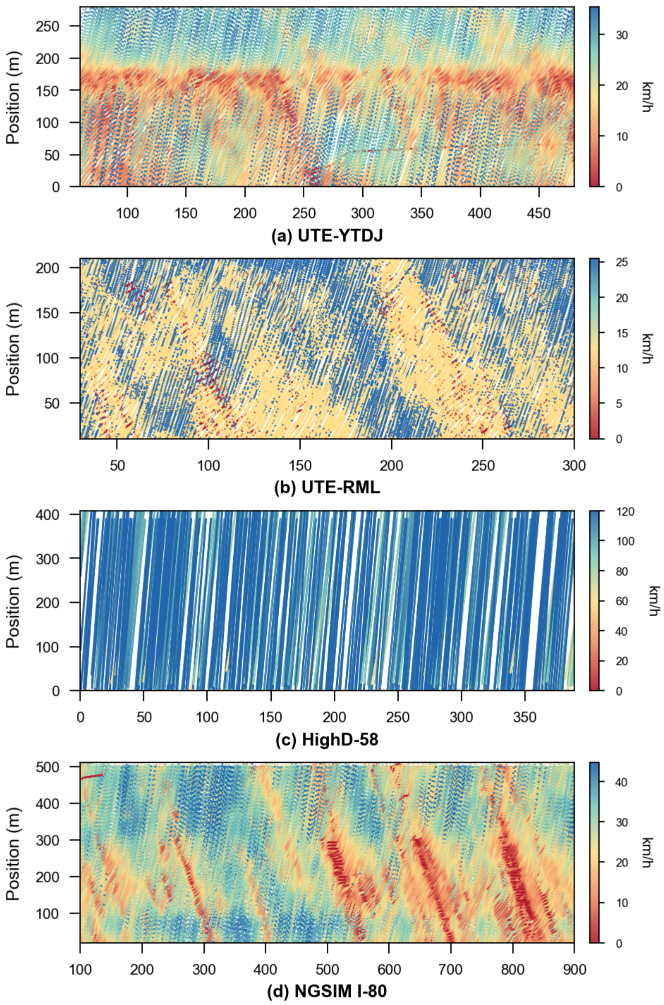


Fig. 3. Observed spacetime speed diagrams synthesized by projecting all analyzed-lane vehicle trajectories onto the time–position plane.

iterations = 80, inner steps = 20, tolerance $5 \times 10^{-4}$. Sensitivity analysis is provided in the companion repository.

## *B. Evaluation Metrics and Protocol*

**Forward consistency** evaluates how the inferred mixture reproduces macroscopic observations. Mixture-implied flow and speed for observation $p$ follow the mixture $q(\rho_p; x_p)$ and $v(\rho_p; x_p)$ (Sec. III; $v$ at $\rho_p = 0$ is the weighted free-flow speed, see (10)). We report $\mathrm{RMSE}_q$, $R_q^2$, $\mathrm{RMSE}_v$, and $R_v^2$, where lower RMSE and higher $R^2$ indicate better fit.

**Transition detection** uses the PED-based bottleneck localization from Sec. IV-C. We report the primary bottleneck site $x^*$, the argmin of PED over $\Omega_Q$, and its transition mean absolute error, T.MAE, against a reference location, together with $\mathrm{PED}_{\min}$, $\Delta\mathrm{PED}$ as the relative drop from the stable-zone baseline, and $H(x^*)$, the Shannon entropy at $x^*$.

**Interpretability and physics consistency.** Phase entropy standard deviation $\sigma_H$ measures how phase-mixing entropy $H(x)$ varies along the segment coordinate $x$; it is not monotonic as a figure of merit. Physical residual (Phys. Res.) is the mean per cell of $(\partial_t \rho + \partial_x q)^2$ on the inferred mixture; lower values indicate better physics consistency.

## *C. Main Results and Interpretation*

Fig. 4 presents the three EM-calibrated prototype FDs. Quality-weighted observations cluster tightly around each triangular prototype across all scenarios, validating the parallelogram sampler's efficacy in isolating quasi-stationary states.

Fig. 5 confirms a tight 1:1 clustering of reconstructed flow and speed with near-zero residuals. Overall, SpinFlow achieves $R_q^2 \geq 0.72$ and $R_v^2 \geq 0.65$ universally (Table III). Performance peaks on the balanced NGSIM I-80 dataset ($R_q^2 = 0.940, R_v^2 = 0.981$), remains robust under heavy urban congestion for YTDJ and RML ($R_q^2 = 0.837$ and $0.744$), and holds steady on sparse HighD data ($R_q^2 = 0.724$). Computationally, the EM algorithm converges within 23–79 iterations, successfully reducing total loss by 96%, even when reaching the 80-iter ceiling on NGSIM. EM convergence curves are also provided in the companion repository.

Fig. 6 and Table II detail inferred phase profiles and PED-based detections. Severe PED drops (94.9–100%) together with $H(x^*) > \ln 2$ are consistent with mixed-phase coexistence at $x^*$. Reported $x^*$ locations agree with documented bottleneck features on YTDJ ($x^* = 66$ m, upstream lane-drop), RML ($x^* = 192$ m, on-ramp mixing edge), and NGSIM I-80 ($x^* = 392$ m, downstream on-ramp boundary). HighD gives the smallest $\mathrm{PED}_{\min} = 0.001$ at $x^* = 88$ m.

TABLE II
BOTTLENECK DETECTION. $x^*$: PRIMARY SITE; $\Delta$PED: DROP VS. STABLE BASELINE; $H(x^*)$: ENTROPY AT $x^*$.

| Scenario | $x^*$ (m) | $\mathrm{PED}_{\min}$ | $\mathrm{PED}_{\mathrm{st}}$ | $\Delta$PED (%) | $H(x^*)$ |
|---|---|---|---|---|---|
| YTDJ (UTE) | 66 | 0.030 | 0.577 | 94.9 | 1.071 |
| RML (UTE) | 192 | 0.019 | 0.499 | 96.3 | 0.912 |
| HighD-58 | 88 | 0.001 | 0.971 | 99.9 | 0.809 |
| NGSIM I-80 | 392 | 0.000 | 0.763 | 100.0 | 0.943 |

## *D. Baseline and Ablation Analysis*

We compare SpinFlow against three heterogeneous baselines representing different modeling paradigms. Table III details their performance and runtime over 5 seeds.

**PWA-CTM (White-box):** Inspired by piecewise-affine approximations of the CTM [34], this original baseline blends three triangular FD zones via an adaptive sigmoid function to explicitly simulate gradual phase coexistence.

**VBGMM+KDE (Grey-box):** Building on GMM-based traffic classification [35], this applies a variational Bayesian Gaussian mixture in $(\rho, q, x)$ space. It incorporates Gaussian KDE for spatial smoothing, enabling continuous phase weight fitting.

**PI-DeepONet (Black-box):** Based on physics-informed operator learning for traffic [36], this Branch–Trunk network is trained with MSE and an LWR continuity loss. It predicts pointwise flow but offers no phase weights or PED signals.

**Baseline results.** SpinFlow achieves the lowest $\mathrm{RMSE}_q$ on YTDJ, HighD, and NGSIM ($p<0.05$ vs. PWA-CTM and VBGMM+KDE; by the same paired $\mathrm{RMSE}_q$ test, PI-DeepONet lags SpinFlow on YTDJ and NGSIM). On near-homogeneous RML, PI-DeepONet attains lower $\mathrm{RMSE}_q$

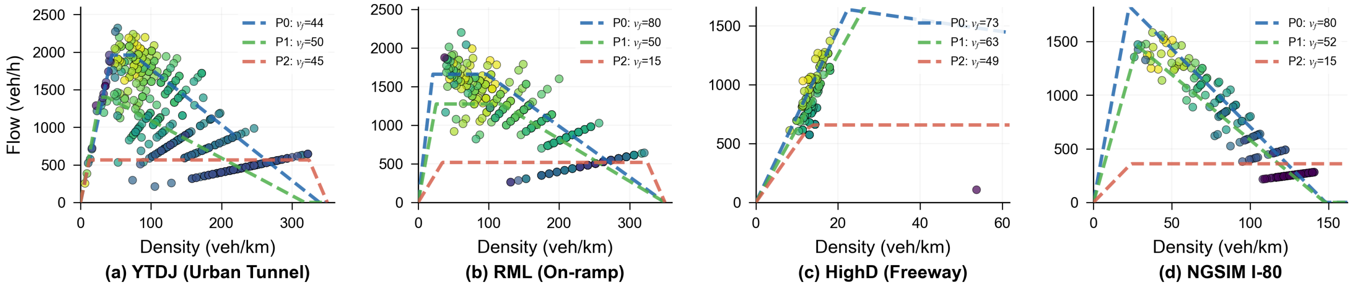


Fig. 4. Fitted three-phase prototype FDs for all four scenarios. Scatter points are parallelogram-sampled observations colored by quality score; dashed curves show the calibrated triangular FD prototypes (F, S, J).

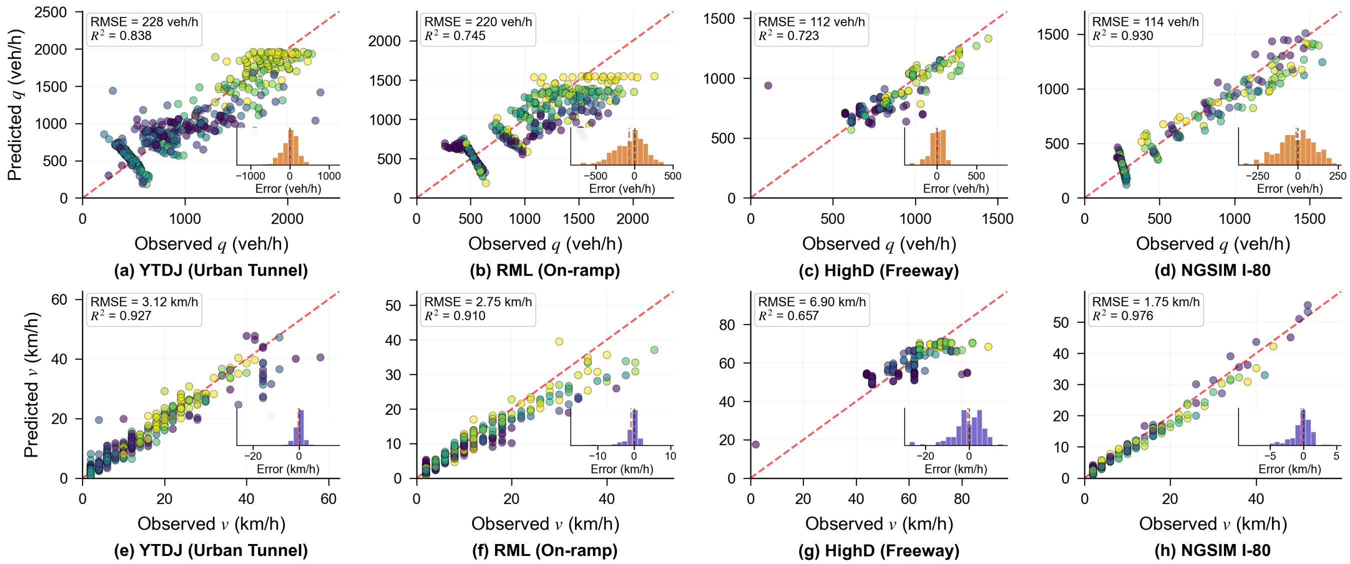


Fig. 5. Forward consistency. Top row (a–d): observed vs. predicted flow $q$ for all four scenarios. Bottom row (e–h): observed vs. predicted speed $v$. Points are colored by spatial position $x$; insets show residual histograms.

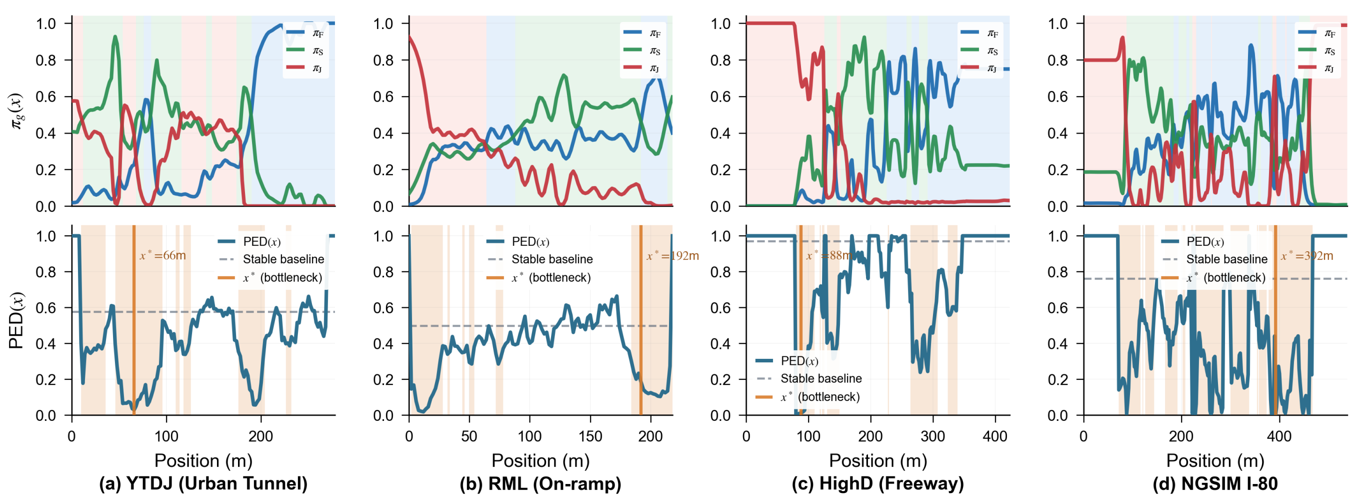


Fig. 6. Phase inference and bottleneck localization for all four scenarios. Top: phase weights $\boldsymbol{\pi}_g(x)$ (blue = F, green = S, pink = J). Bottom: PED($x$) with non-equilibrium zones (red fill), stable baseline (dashed), and $x^*$ (red line).

(201.8 vs. 219.8 veh/h) but outputs neither calibrated $\boldsymbol{\pi}(x)$ nor PED. On YTDJ, PWA-CTM is constrained when co-existence is gradual (327.6 veh/h), whereas PI-DeepONet degrades strongly ($R_v^2$=−0.42) under strong LWR continuity regularization. On HighD, SpinFlow attains T.MAE=0.0 m with RMSE$_q$=111.9 veh/h; PI-DeepONet yields comparable flow RMSE on average ($p$=0.055) but larger variance across seeds. On NGSIM, SpinFlow reaches $R_q^2$=0.940.

TABLE III

ACCURACY AND EFFICIENCY ACROSS FOUR SCENARIOS. BEST RESULTS IN BOLD; *: $p<0.05$ VS. SPINFLOW (PAIRED $t$-TEST ON RMSE$_q$).

| Scenario | Model | RMSE$_q$ ↓ | $R_q^2$ ↑ | RMSE$_v$ ↓ | $R_v^2$ ↑ | T.MAE (m) ↓ | Phys. Res. ↓ | $\sigma_H$ ↔ | RT (s) ↓ | Iter ↓ | $\pi(x)$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| UTE-YTDJ (Bottleneck) | SpinFlow | **228.1**±0.0 | **0.837** | **3.14**±0.00 | **0.926** | 34.0±0.0 | 5.29e-4 | 0.355 | 3.61±0.66 | 59 | Yes |
| | PWA-CTM* | 327.6±0.3 | 0.664 | 3.42±0.00 | 0.913 | 28.0±0.0 | **2.84e-4** | 0.238 | 0.07±0.00 | **1** | Partial |
| | VBGMM+KDE* | 277.8±0.0 | 0.759 | 3.83±0.00 | 0.890 | 114.0±0.0 | 4.63e-4 | 0.282 | 0.67±0.78 | 22 | Partial |
| | PI-DeepONet* | 361.9±29.9 | 0.588 | 13.69±1.80 | −0.418 | **0.0**±0.0 | 3.61e-4 | 0.000 | 5.25±2.38 | 1420 | No |
| | No Comp.* | 241.0±0.1 | 0.818 | 3.21±0.00 | 0.917 | 34.0±0.0 | 4.93e-4 | 0.234 | 1.97±0.09 | 36 | Yes |
| | Unit Norm* | 239.8±1.3 | 0.820 | 3.30±0.03 | 0.919 | 34.0±0.0 | 4.53e-4 | 0.273 | 1.60±0.32 | 25 | Yes |
| | No Phys.* | 230.6±0.0 | 0.834 | 3.26±0.00 | 0.921 | 34.0±0.0 | 5.12e-4 | 0.350 | 2.25±0.18 | 42 | Yes |
| | Single FD* | 372.8±0.0 | 0.565 | 3.90±0.00 | 0.886 | **0.0**±0.0 | 3.28e-4 | 0.000 | **0.01**±0.00 | **1** | Yes |
| UTE-RML (On-ramp) | SpinFlow | 219.8±0.0 | 0.744 | 2.76±0.00 | 0.909 | 166.0±0.0 | 2.38e-4 | 0.184 | 2.75±0.06 | 75 | Yes |
| | PWA-CTM* | 224.5±0.1 | 0.733 | 1.99±0.01 | 0.953 | 122.0±0.0 | 3.63e-4 | 0.245 | **0.05**±0.00 | **1** | Partial |
| | VBGMM+KDE* | 381.7±0.1 | 0.228 | 4.82±0.01 | 0.723 | 166.0±0.0 | **2.09e-4** | 0.223 | 0.45±0.06 | 46 | Partial |
| | PI-DeepONet* | **201.8**±1.2 | **0.784** | **1.89**±0.01 | **0.957** | **98.0**±0.0 | 4.30e-4 | 0.271 | 4.50±1.03 | 1321 | No |
| | No Comp.* | 234.1±0.0 | 0.710 | 3.03±0.00 | 0.890 | 166.0±0.0 | 2.22e-4 | 0.110 | 0.78±0.01 | 17 | Yes |
| | Unit Norm* | 230.9±0.5 | 0.718 | 3.03±0.01 | 0.890 | 166.0±0.0 | 2.16e-4 | 0.127 | 1.59±0.33 | 40 | Yes |
| | No Phys.* | 231.0±0.0 | 0.718 | 3.04±0.00 | 0.890 | 166.0±0.0 | 2.25e-4 | 0.151 | 1.51±0.03 | 39 | Yes |
| | Single FD* | 224.5±0.0 | 0.733 | 1.95±0.00 | 0.955 | **98.0**±0.0 | 3.18e-4 | 0.000 | **0.02**±0.00 | **1** | Yes |
| HighD-58 (Segment) | SpinFlow | **111.9**±0.0 | **0.724** | **6.89**±0.00 | **0.658** | **0.0**±0.0 | 1.25e-5 | 0.279 | 0.70±0.02 | 23 | Yes |
| | PWA-CTM* | 133.5±0.7 | 0.608 | 8.13±0.06 | 0.525 | 127.8±58.0 | 1.33e-5 | 0.208 | 0.05±0.00 | **1** | Partial |
| | VBGMM+KDE* | 138.4±0.0 | 0.578 | 8.55±0.00 | 0.474 | 6.0±0.0 | 1.15e-5 | 0.434 | 0.27±0.10 | 36 | Partial |
| | PI-DeepONet | 143.5±25.6 | 0.535 | 9.90±1.14 | 0.287 | 119.8±0.0 | 1.40e-5 | 0.013 | 6.56±0.05 | 2000 | No |
| | No Comp.* | 136.0±0.7 | 0.593 | 8.00±0.03 | 0.539 | 6.0±0.0 | 1.13e-5 | 0.214 | 0.68±0.08 | 21 | Yes |
| | Unit Norm* | 118.8±0.1 | 0.689 | 7.39±0.01 | 0.607 | 2.0±0.0 | 1.22e-5 | 0.218 | 0.66±0.19 | 21 | Yes |
| | No Phys.* | 112.7±0.0 | 0.720 | 6.93±0.00 | 0.655 | 2.0±0.0 | 1.25e-5 | 0.270 | 2.25±0.03 | 80 | Yes |
| | Single FD* | 171.0±0.0 | 0.355 | 10.97±0.00 | 0.134 | 119.8±0.0 | **1.09e-5** | 0.000 | **0.02**±0.00 | **1** | Yes |
| NGSIM I-80 (Bottleneck) | SpinFlow | **105.7**±1.3 | **0.940** | **1.53**±0.05 | **0.981** | 308.0±0.0 | 7.30e-5 | 0.298 | 3.94±0.08 | 79 | Yes |
| | PWA-CTM* | 123.3±2.4 | 0.918 | 1.59±0.03 | 0.980 | **201.2**±1.1 | 8.14e-5 | 0.221 | 0.06±0.00 | **1** | Partial |
| | VBGMM+KDE* | 270.3±0.1 | 0.606 | 5.45±0.00 | 0.763 | 300.0±0.0 | 1.43e-4 | 0.299 | 0.33±0.07 | 31 | Partial |
| | PI-DeepONet* | 118.9±4.7 | 0.924 | 1.76±0.15 | 0.975 | 304.0±0.0 | 1.10e-4 | 0.170 | 5.30±2.13 | 1560 | No |
| | No Comp.* | 131.4±0.0 | 0.907 | 2.08±0.00 | 0.966 | 320.0±0.0 | **5.23e-5** | 0.255 | 1.32±0.05 | 23 | Yes |
| | Unit Norm* | 130.9±1.3 | 0.908 | 2.11±0.03 | 0.964 | 308.0±0.0 | 5.65e-5 | 0.229 | 2.20±1.66 | 41 | Yes |
| | No Phys.* | 130.3±0.2 | 0.908 | 2.07±0.01 | 0.966 | 308.0±0.0 | 5.47e-5 | 0.320 | 1.05±0.21 | 16 | Yes |
| | Single FD* | 121.9±0.0 | 0.920 | 1.55±0.00 | 0.978 | 304.0±0.0 | 7.85e-5 | 0.000 | **0.02**±0.00 | **1** | Yes |

Mean±std over 5 seeds. Units: RMSE$_q$ (veh/h), RMSE$_v$ (km/h), Phys. Res. $(\mathrm{veh}/(\mathrm{m}\cdot\mathrm{s}))^2$. $\pi(x)$: Yes = calibrated, Partial = proxy, No = N/A.

**Ablation results.** Four ablations remove one SpinFlow mechanism at a time: **No Comp.** disables the competition penalty ($h_S{=}s_x$); **Unit Norm** projects $\mathbf{s}$ onto the unit sphere, eliminating Phase Unfolding; **No Phys.** sets $\lambda_{\text{phys}}{=}0$; **Single FD** reduces the model to standard single-prototype LWR.

Removing the competition mapping raises RMSE$_q$ by 5.7–24.3 % across all scenarios ($p{<}0.05$). On YTDJ, $\sigma_H$ decreases from 0.355 to 0.234, consistent with weaker along-$x$ variability in mixing entropy when the three-phase competition pathway is disabled; $\sigma_H$ is reported descriptively and is not used as a scalar objective. Synchronized weight $\pi_S$ is suppressed, yielding an effective free–jam partition. Constraining the spin to unit norm increases RMSE$_q$ by 5–24 %, with the largest gap on NGSIM under extreme densities, indicating that Phase Unfolding helps localize sharp phase dominance. Omitting the physics prior increases RMSE$_q$ most on NGSIM (23.2 %) and only modestly on YTDJ (+1.1 %); on sparse HighD the effect is negligible as LWR gradients shrink at convergence. Single FD sharply raises RMSE$_q$ (63.4 % on YTDJ, 52.8 % on HighD) and ties transitions to a density-gradient heuristic, yielding scattered T.MAE (0, 98, 120, and 304 m) when gradient peaks misalign with the reference bottleneck. PED sidesteps that coupling—notably T.MAE=0.0 m on HighD—without hand-tuned thresholds.

In summary, SpinFlow strikes a practical balance among accuracy, interpretability, and computational efficiency. It achieves strong in-sample flow fit ($R_q^2{\geq}0.72$), yields structured phase maps ($\sigma_H{>}0$), and supports PED-based bottleneck localization. Per scenario on a single CPU, training converges within 0.7–3.9 s and is up to 9× faster than PI-DeepONet (4.5–6.6 s); conversely, analytic calibrations run in under $<0.1$ s but do not perform explicit phase inference.

## VI. CONCLUSION

SpinFlow translates Kerner's three-phase theory into a computable framework, partly addressing the two critical gaps identified in Sec. I. First, bridging microscopic mechanisms with computable states, the competitive-equilibrium mapping lets synchronized flow emerge without pre-specified static boundaries, while physics-regularized EM and Phase Unfolding yield interpretable phase maps. Second, replacing rigid empirical thresholds, PED quantifies model–data structural alignment and localizes coexistence in a topology-free manner. Across four real-world datasets, SpinFlow achieves $R_q^2$ up to

0.940 and PED drops of 94.9–100 %; relative to heterogeneous baselines, it delivers calibrated $\boldsymbol{\pi}(x)$ with favorable $\mathrm{RMSE}_q$ except against PI-DeepONet on RML. SpinFlow thereby offers an ATM-oriented precursor signal grounded in phase structure rather than fixed thresholds alone. This motivates viewing traffic dynamics as spatial phase competition with emergent signatures, complementary to hydrodynamic models.

Limitations remain. First, gains over a single-phase FD shrink to $\sim 2\,\%$ on near-homogeneous scenarios (e.g. RML), suggesting mild redundancy when multi-phase structure is weak. Second, quasi-stationary parallelogram sampling can yield spurious EM artifacts under rapidly evolving disturbances. Third, validation is confined to 1-D ordered segments: (i) lane-changing and lateral competition are not explicitly represented; (ii) network-scale merges and diverges, with topology-aware generalization, remain open; (iii) online deployment calls for latency-robust incremental updates and closed-loop stability guarantees. Future work will target online incremental EM for edge-side real-time spin-field updates, multi-lane spin fields for cross-lane phase competition, graph-aware kernels with inter-segment coupling, and closed-loop integration with Vehicle-to-Everything (V2X) and Connected and Automated Vehicle (CAV) systems in which PED can trigger variable speed limits or ramp metering—advancing phase inference toward preventive ATM.

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